



A vision transformer machine learning model for COVID-19 diagnosis using chest X-ray images

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ABSTRACT

This study leverages machine learning to enhance the diagnostic accuracy of COVID-19 using chest X-rays. The study evaluates various architectures, including efficient neural networks (EfficientNet), multiscale vision transformers (MViT), efficient vision transformers (EfficientViT), and vision transformers (ViT), against a comprehensive open-source dataset comprising 3616 COVID-19, 6012 lung opacity, 10192 normal, and 1345 viral pneumonia images. The analysis, focusing on loss functions and evaluation metrics, demonstrates distinct performance variations among these models. Notably, multiscale models like MViT and EfficientNet tend towards overfitting. Conversely, our vision transformer model, innovatively fine-tuned (FT) on the encoder blocks, exhibits superior accuracy: 95.79% in four-class, 99.57% in three-class, and similarly high performance in binary classifications, along with a recall of 98.58%, precision of 98.87%, F1 score of 98.73%, specificity of 99.76%, and area under the receiver operating characteristic (ROC) curve (AUC) of 0.9993. The study confirms the vision transformer model's efficacy through rigorous validation using quantitative metrics and visualization techniques and illustrates its superiority over conventional models. The innovative fine-tuning method applied to vision transformers presents a significant advancement in medical image analysis, offering a promising avenue for improving the accuracy and reliability of COVID-19 diagnosis from chest X-ray images.

1. Introduction

Chest X-rays are a crucial tool for diagnosing various illnesses. However, their current use faces limitations in efficiency, particularly during critical situations like pandemics. The delay of 1–2 days between scans and diagnosis can significantly impact patient outcomes. This challenge is particularly evident with diseases like COVID-19, which has affected over 772 million people globally, with over 7 million fatalities [1]. The COVID-19 pandemic resulted in a significant economic burden, with a 3.3 trillion dollar deficit in the US for 2020 and a peak unemployment rate of 14.7% [2,3]. Medical institutions faced surging demand, leading to longer wait times and exacerbated disparities in access to care. Studies in [3,4] found a concerning 117% increase in wait time disparities. To address these challenges, a critical re-evaluation of current diagnostic approaches is essential. Embracing innovative solutions and integrating technology are key to improving efficiency, minimizing delays, and optimizing healthcare outcomes.

The critical need for speed and accuracy in medical diagnosis has driven researchers to explore the transformative potential of artificial

intelligence (AI), particularly advanced deep learning networks. These early models achieved promising levels of accuracy in medical image analysis, paving the way for advancements in diagnostic classification. For instance, authors in [5] got an accuracy of 95.1% with testing K-nearest neighbor (KNN), support vector machine (SVM) and convolutional neural networks (CNN) method on 196 COVID-19 and normal chest X-ray images. Xiao et al. [6] designed the PAM-DenseNet model, which comprised of a hybrid of residual-concatenated dense blocks [7] with a parallelly arranged channel attention map and spatial attention map blocks, to obtain an accuracy of 94.29%, a precision of 93.75%, a sensitivity of 95.74%, and a specificity of 96.77% on classifying patient computed tomography scan (CT) dataset consisting of COVID-19, normal, and pneumonia images. Wu et al. [8] proposed the COVID-AL network architecture to predict segmented lung CT images (304 for COVID-19, 316 for pneumonia, and 342 for normal) with an accuracy of 86.6%. Panahi et al. [9] devised the fast COVID-19 detector structure, which composed of hybrid and parallel depth-wise separable convolution layers, achieving an accuracy of 96% on binary

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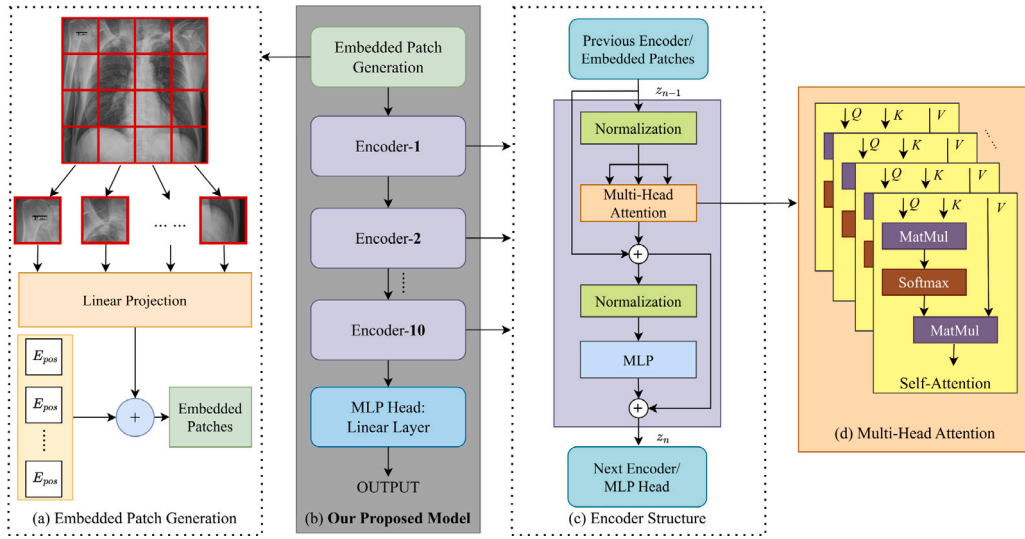


Fig. 1. The structure of our proposed model.

classification with 435 COVID-19 and 505 non-COVID-19 chest X-ray images. Utilizing CNN as the bottleneck and KNN as the bottom classifier with a novel evolutionary strategy, Jalali et al. [10] achieved an accuracy of 98.57% on predicting 192 COVID-19 and 192 non-COVID-19 chest X-ray images. Rayan and Alaerjan [11] implemented the bidirectional long short term memory (Bi-LSTM) model to train with preprocessed 181 COVID-19 and 46 normal chest X-ray images and get 93.48% accuracy. Chetoui and Akhloufi [12] tested vision transformer (ViT) models, ViT-B16, ViT-B32, and ViT-L32, with 7598 COVID-19, 8552 normal, and 5674 Pneumonia CXR images, resulting in a 96% classification accuracy.

Our further analysis reveals that most existing COVID-19 detection models are trained on datasets with limitations, such as a lack of image diversity. Additionally, few studies have explored transformer-based architectures [13]. While pioneering attempts achieved commendable accuracy, they often suffered from overfitting due to inadequate management or limited fine-tuning, such as adjusting encoder blocks within the transformer model. Conversely, simplifying the vision transformer by reducing encoder blocks could potentially mitigate overfitting and decrease complexity. Inspired by the success of EfficientNet [14] in COVID-19 diagnosis [15–17] and the Vision Transformer's strength in image classification, we conducted meticulous experiments with both CNN and ViT models. Our work introduces a finely tuned ViT model that aims to reduce training time without sacrificing accuracy, making it a promising solution for efficient COVID-19 detection. Our key contributions are summarized as follows:

1. We summarize the existing machine learning models used in COVID-19 lung image classification.
2. We analyze the structure of the experimented models with a focus on the vision transformer.
3. We conduct experiments with various model structures on a COVID-19 chest X-ray dataset and analyze their performance.
4. We propose our fine-tuned vision transformer model from the early experiment findings.
5. We evaluate the performance of our proposed model using standard metrics and visualization techniques to quantify its effectiveness and efficiency in COVID-19 detection from chest X-ray images.

The remainder of this paper is organized as follows: We examine existing models and fine-tuning strategies in Section 2. Section 3 details the experimental setup, including the dataset, preprocessing, model

configurations, and evaluation metrics. Section 4 presents the key findings, including dataset biases, model strengths and limitations, and the broader implications for revolutionizing medical diagnostics. Section 5 concludes the paper and Section 6 discusses potential avenues for future work related to this research.

2. Methodology

This section first introduces the proposed model architectures used in our experiments, as illustrated in Fig. 1. We then delve into the design and fine-tuning technique of our focus model, the vision transformer, followed by an analysis on the structure of the other experimented models for comparison.

2.1. Proposed model

Vision transformer (ViT) models [13] stand apart from CNNs due to their unique mathematical architecture and foundational blocks. The primary distinction between the ViT structure and traditional CNN architecture lies in ViT's extensive global receptive field, where the receptive field is the region that computer vision models can understand [18]. By leveraging a global receptive field, ViT can capture relationships between distant (non-adjacent) image patches, potentially leading to superior performance in image classification tasks.

2.1.1. Embedded patch generation

To transform the input for further processing, the input image is meticulously partitioned into patches of size P . Undergone a rigorous linear projection, these patches are flattened and culminate in the formation of patch embeddings. To preserve the spatial relationships between these patches, the model dynamically generates positional embeddings E_{pos} , which are treated as learnable parameters during training. The model subsequently merges the linearly projected patch embeddings with these positional embeddings, and the resulting sum is diligently fed into the encoder blocks, preceded by a crucial normalization layer. This process is illustrated in Fig. 1(a) and can be further explained by the following expression:

$$E_{patches} = [x_{class}, x_p^1 W, x_p^2 W, \dots, x_p^n W] + E_{pos} \quad (1)$$

where W represents the linear projection metrics, E_{pos} is the position embedding and x_{class} denotes class token. x_{class} is a randomly initialized array used for storing accumulated information across the entire image and also feeding into the last layer for final classification.

2.1.2. Encoder block

To capture inner patterns of the dataset in high dimensions, encoder blocks are introduced. These transformations are achieved by a linear layer within each block. This layer transforms the input into three outputs: query (Q), key (K), and value (V), by multiplying the input with three separate matrices: W_Q , W_K , and W_V . Subsequently, these outputs are processed through a similarity-learning function, enabling each patch to learn its relation to other patches. This process, known as self-attention (SA), allows a patch to consider information from all other patches, thus capturing global dependencies (as shown in Fig. 1(d)). Furthermore, the encoder block employs multiple parallel self-attention mechanisms. These parallel operations are concatenated and then multiplied by a learnable matrix W_{proj} . This entire procedure is referred to as multi-headed self-attention (MSA), which enhances the model's ability to focus on various parts of the input simultaneously, thereby improving its representational power. This procedure is realized by:

$$z_0 = E_{patches} \quad (2)$$

$$\text{Query (Q)} = z_{n-1} \cdot W_Q$$

$$\text{Key (K)} = z_{n-1} \cdot W_K \quad (3)$$

$$\text{Value (V)} = z_{n-1} \cdot W_V$$

$$\text{SA} = \text{Softmax}\left(\frac{Q \cdot K^T}{ES}\right) \cdot V \quad (4)$$

$$\text{MSA} = [\text{SA}_1, \text{SA}_2, \text{SA}_3, \dots, \text{SA}_n] \cdot W_{proj} \quad (5)$$

where z_{n-1} is the input of the encoder block. ES stands for embedding size, which is the size of activations after patches are linearly projected. In the implemented models, its size equals $\text{patch_size} \times \text{patch_size} \times \text{image_channels}$. At the end of MSA, a residual connection is applied:

$$z' = \text{MSA} + z_{n-1} \quad (6)$$

After MSA, a multilayer perceptron (MLP) block which contains two linear layers is applied to learn the local information and the complexity. It also has a residual connection at the output:

$$z_n = \text{MLP}(\text{LN}(z')) + z' \quad (7)$$

where z_n is the output of the encoder block, and LN stands for layer normalization. The encoder structure is demonstrated at Fig. 1(c)

The output of MLP is then fed into the next encoder block, except the last one. At the bottom, the output of MLP goes inside a linear layer (called MLP head) in the pre-trained model, where the number of output nodes of the linear layer is adjusted to the same size (or number) of prediction classes. The overall view of our proposed ViT model, configuring with 16 by 16 patch divisions and 10 encoders, is introduced in Algorithm 1 and Fig. 1(b).

2.2. Weight decay

Since overfitting often occurs in model training and affects the final prediction precision significantly, it is vital to minimize its influence [19]. Inspired by [20] and intuitive work [21], we insert the weight decay penalty value in parameter update rules to mitigate the observed overfitting in experiments (as shown in the loss Fig. 3). Weight decay, or L_2 regularization, is a crucial method that we utilize for fine-tuning our models. It adds a penalty to the loss to avoid the weights having an upheaval change during updating:

$$\|\mathcal{W}\|^2 = \mathbf{w}_1^2 + \mathbf{w}_2^2 + \dots + \mathbf{w}_n^2 \quad (8)$$

$$\text{Loss}_{updated}(\mathcal{W}) = \text{Loss}_{original}(\mathcal{W}) + \lambda \|\mathcal{W}\|^2$$

where $\mathcal{W}$ is the integration of all the trainable parameters inside the model. λ , the penalty constant, which reflects the influence of the sum of square values of parameters on the final loss outcome, serves as one of the hyperparameters in our experiment. $\text{Loss}_{updated}$ is fed into the

Algorithm 1: Vision Transformer Workflow.

```

1 Input: Image  $I$ ,  $E_{pos}$ ,  $x_{class}$ ,  $\alpha_{mlp\_ratio}$ ,  $n$  (number of classes)
2  $x_{patches} \leftarrow \text{Flatten}(\text{Convolution}(I, k, s))$ 
3  $x = [x_{class}; x_{patches}] + E_{pos}$ 
4  $d = \left(\frac{\text{image size}}{\text{patch size}}\right)^2$ 
5 Output:  $\text{LinearLayer}(x[x_{class}], d, n)$ 
6 for ( $0 \leq i \leq \text{number of encoders}$ ) do
7    $x_{norm1} \leftarrow \text{Normalize}(x)$ 
8   MHA (Attention):
9    $q, k, v \leftarrow \text{LinearProject}(x_{norm}, d, 3 \times d)$ 
10   $x_{attn} \leftarrow \text{Softmax}(q, k) \cdot v$ 
11   $x_{atten} \leftarrow \text{LinearProject}(x_{attn}, 3 \times d, d)$ 
12  Scaling and residual connection:
13   $x' \leftarrow \text{LayerScaling}(x_{atten}) + x$ 
14  MLP:
15   $x_{norm2} \leftarrow \text{Normalize}(x')$ 
16   $x_{fc1} \leftarrow \text{Activation}_{GELU}(\text{LinearLayer}(x_{norm2}, d, d \times \alpha_{mlp\_ratio}))$ 
17   $x_{fc2} \leftarrow \text{LinearLayer}(x_{fc1}, d \times \alpha_{mlp\_ratio}, d)$ 
18   $x \leftarrow \text{LayerScaling}(x_{fc2}) + x'$ 
19 return  $\text{LinearLayer}(x[x_{class}], d, n)$ 

```

optimizer for updating weights in the next iteration. This implementation makes parameter updates not only subject to gradients of loss but also consider the weights' values, which has proved to be a critical technique to prevent transformers' overfitting in our experiments.

2.3. Experimented models

In our experiment, we employ four pre-trained model architectures for transfer learning on the COVID-19 chest X-ray dataset: EfficientNet [22], enhanced multiscale vision transformer (MViT) [23], EfficientViT [24], and vision transformer (ViT) [13]. EfficientNet, known for its computational efficiency and robustness, is constructed using a series of inverted residual blocks (MBConv blocks) [25], with varying layer configurations. These configurations are optimized through a grid search, limiting depth, width, and resolution scaling to a constant factor, alpha. These varying blocks contribute to its multiscale learning structure, which enables it to capture more details from the input image.

Building on the findings discussed in Section 1, we hypothesize that the scaled structure of EfficientNet may also benefit transformer-based models in extracting features from the COVID-19 chest X-ray dataset. The analysis of each experimented transformer model is as follows. MViT realizes multiscale learning by learning on reduced-resolution tensors in each level of the encoder. It incorporates a pooling layer before the query, key, and value tensors. EfficientViT, similar to MViT, incorporates a multiscale learning pyramid structure through the integration of depthwise convolution, enabling the modification of tensor resolution. Additionally, EfficientViT replaces the Softmax function in the similarity function with the ReLU activation function [26] and employs several MBConv blocks ahead of its transformer blocks, significantly improving hardware computational efficiency. Each of these state-of-the-art models has demonstrated exceptional performance in classifying complex datasets, such as ImageNet-1k/ImageNet-21k.

3. Experiments

3.1. Dataset and preprocess

The COVID-19 chest X-ray dataset used in this study was sourced from the publicly available COVID-19 Radiography Database [27–29]. This dataset comprises four distinct classes: 3616 COVID-19 positive

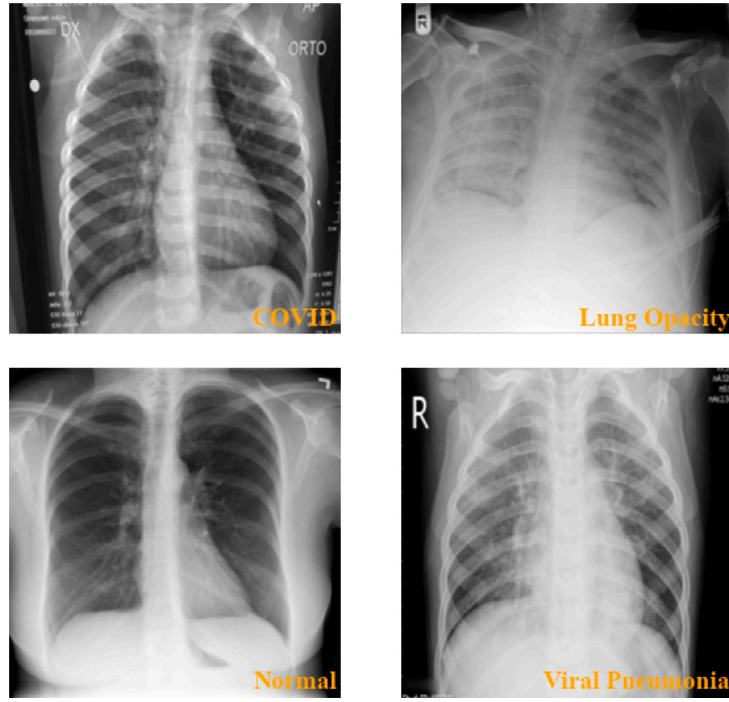


Fig. 2. COVID-19 chest X-ray dataset.

cases, 10,192 normal cases, 6012 lung opacity (non-COVID lung infection) cases, and 1345 viral pneumonia images, with an initial resolution of 299 by 299, as illustrated in Fig. 2. In our experiments, the dataset is fully shuffled and split into 80% for training, 10% for validation, and 10% for testing.

To address potential data imbalances and enhance training efficiency, we preprocess the images in several ways. First, we normalize the original pixel values from [0, 255] to [0.0, 1.0]. This normalization step facilitates the training process by ensuring all pixel values are on a similar scale. Second, we resize the images to 224 by 224 pixels to match the input requirements of the pre-trained models we employ for transfer learning [30]. Inspired by the work in [28], which investigated various preprocessing techniques for chest X-ray images with CNNs, we conduct additional experiments with the following preprocessing methods: barely normalized images, (partially) segmented images based on provided masks in the dataset, and gamma-corrected images with different levels of constant settings.

3.2. Model configuration

We employ transfer learning with pre-trained models on the COVID-19 chest X-ray dataset. Through fine-tuning on preprocessing methods (bare normalization, segmentation and gamma correction), learning rate (range [1e-3, 1e-6]), batch size (within 16, 32, 64, 128), and weight decay (range [1e-1, 1e-6]), we obtain optimized settings (as reported in Table 1) and corresponding models. Additionally, the cross-entropy function [31] served as the loss function, and AdamW/Adam [32,33] are utilized as optimizers for experimented models. Then, observing the superior performance of ViT-B16 compared with others in the early stage of experiments, we further fine-tuned it by setting different numbers of encoder blocks inside. The configurations and results on different numbers of block settings are shown in Table 2.

3.3. Evaluation methodology

To evaluate our model's performance on the four-class classification task, we utilize key metrics such as accuracy, recall (or sensitivity/true positive rate), precision, and F1 score. We additionally generate the

receiver operating characteristic (ROC) curve and calculate the area under the ROC curve (AUC) values. All of these metrics are grounded in the fundamental concepts of true positive (TP), true negative (TN), false positive (FP), and false negative (FN). TP represents instances correctly classified as positive, TN indicates instances correctly classified as negative, FP signifies instances incorrectly predicted as positive, and FN represents instances incorrectly predicted as negative. These metrics collectively offer a comprehensive assessment of a classification model's performance, considering the effectiveness of COVID-19 prediction.

$$\text{Accuracy (A)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$\text{Recall (R)} = \frac{TP}{TP + FN} \quad (10)$$

$$\text{Precision (P)} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (12)$$

$$\text{F1 Score (F1)} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (13)$$

The ROC curve is constructed by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings, which determine the outcome of predictions. An ideal model's ROC curve would appear as a rectangle, achieving an AUC value of 1, signifying perfect classification. The ROC curve essentially measures the model's ability to discriminate between positive and negative cases across different thresholds, reflecting the balance between sensitivity and specificity.

$$\begin{aligned} \text{TPR} &= \frac{TP}{TP + FN} \\ \text{FPR} &= \frac{FP}{FP + TN} \\ \text{ROC}[i] &= \frac{\text{TPR}[i]}{\text{FPR}[i]} \end{aligned} \quad (14)$$

4. Result analysis

4.1. Train loss analysis

To assess the training effectiveness of the experimented models, we monitor the loss function on both the training and validation datasets.

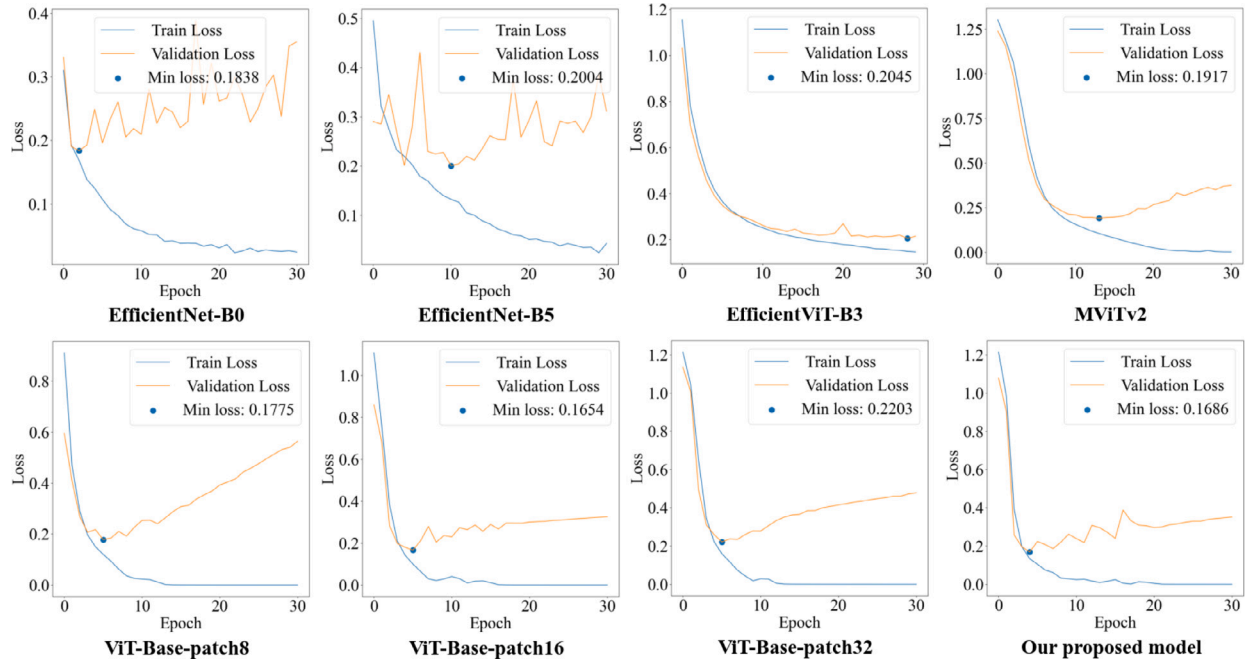


Fig. 3. Train and validation losses during training process.

Table 1

Optimized training configurations after fine-tuning.

Model	Preprocess	Learning rate	Batch size	Weight decay
EfficientNet-B0	Gamma correction (-0.8)	1.00e-03	16	1.00e-05
EfficientNet-B5	Bare normalization	1.00e-03	16	1.00e-05
EfficientViT-B3	Bare normalization	2.00e-06	32	1.00e-02
MViTv2	Bare normalization	2.00e-06	16	1.00e-02
ViT-Base-patch8	Bare normalization	2.00e-06	16	1.00e-01
ViT-Base-patch16	Bare normalization	1.00e-05	64	1.00e-02
ViT-Base-patch32	Bare normalization	1.00e-05	64	1.00e-01
Proposed model	Bare normalization	1.00e-05	64	5.00e-03

Table 2

Training configuration and test results after fine-tuning on ViT-16 (batch size: 64, learning rate: 1e-5, preprocessed with bare normalization).

Number of encoder blocks	Weight decay	Accuracy (%)	Recall (%)	Precision (%)	F1 Score (%)
5	1.00e-02	94.37	98.31	98.58	98.44
7	1.00e-02	95.50	99.15	98.87	99.01
8	1.00e-02	95.55	98.87	98.59	98.73
9	1.00e-03	95.41	98.87	99.15	99.01
10 (proposed)	5.00e-03	95.79	98.58	98.86	98.72
11	1.00e-02	95.64	99.15	98.87	99.01
12 (original)	1.00e-02	95.22	98.31	98.86	98.58

The loss curves for each model (as depicted in Fig. 3) are obtained through the fine-tuning process detailed in Section 3.2. Notably, weight decay plays a crucial role in achieving optimal performance.

Convergence Rate: Under the optimized settings reported in Table 1, EfficientNet-B0, ViT-Base-patch8, ViT-Base-patch16, and ViT-Base-patch32 achieve minimum loss within 10 iterations, demonstrating their superior convergence speed. MViTv2 and EfficientNet-B5 achieve minimum loss between 10 to 15 iterations, whereas EfficientViT-B3 requires nearly 30 iterations to reach its minimum.

Overfitting: We employ the cross-entropy loss function and implement early stopping with a patience of 30 epochs to mitigate overfitting. As observed in the training and validation loss curves depicted in Fig. 3, the extent of overfitting varies across models. Notably, EfficientViT-B3 exhibits a stable fitting behavior, while EfficientNet models demonstrates a tendency towards relatively unstable overfitting on our experimental dataset.

Our experimented models demonstrate relatively promising results after fine-tuning, with validation loss values typically fluctuating in the range of 0 to 0.2. Trading off with the convergence rate, overfitting, and loss value, we select the ViT-Base-patch16 structure for further tuning, where our proposed model is generated. Table 2 shows the corresponding configurations and results. Compared to other models, our proposed model achieves lower training loss and higher accuracy, although a slight overfitting gap is still present. A detailed analysis of the prediction performance is provided in the next subsection.

4.2. Test performance analysis

After the training process, we proceed with testing on the test set, evaluating the results based on the specified criteria, and finally comparing them with existing models. Firstly, we observe that the training loss accurately reflects the achieved accuracy when the experimental

Table 3
Fine-tuned experimented model's result comparison.

Model	Accuracy (%)	Recall (%)	Precision (%)	F1 score (%)	Trainable parameters (million)	Training time per epoch (s)
EfficientNet-B0	94.65	98.58	98.31	98.44	4.34	90.31
EfficientNet-B5	93.94	99.43	97.23	98.32	28.35	286.26
EfficientViT-B3	94.51	98.59	98.03	98.31	46.09	276.45
MViT2	94.41	96.88	97.71	97.29	50.71	492.49
ViT-Base-patch8	95.41	99.15	98.59	98.87	85.81	1652.94
ViT-Base-patch16	95.22	98.31	98.86	98.58	85.81	313.94
ViT-Base-patch32	93.71	98.29	97.47	97.88	87.46	89.77
Proposed model	95.79	98.58	98.87	98.73	71.62	264.79

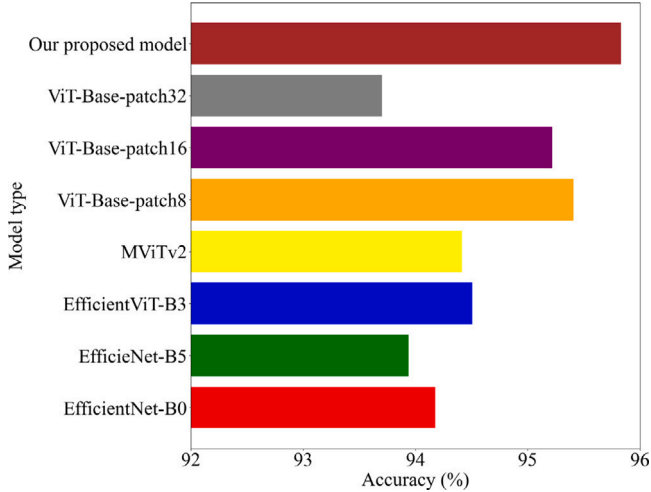


Fig. 4. Experimented models performance comparison (four-class classification).

models reach a high accuracy score (around 93% to 96%) as shown in Table 3 and Fig. 4. However, this is also balanced by the training time per epoch; models with higher accuracy often require more resources to train as the complexity of the model increases. For instance, the ViT-Base-patch8 has excellent performance with accuracy (95.41%), recall (99.15%), precision (98.59%), and F1 score (98.87%) among fine-tuned models. However, with a trainable parameter of 85.81 M, the training time taken in each iteration is several times longer than in other models (6.2 times longer in comparison to our model), as shown in Table 4. Although it has a similar number of trainable parameters ($(224/8)^2 \times 8^2 \times 3$) as ViT-Base-patch16's ($(224/16)^2 \times 16^2 \times 3$), ViT-Base-patch8 has more tokens (patches) ($(224/8)^2$) which exhibits more computational cost than flattened features (8^2). EfficientNetB5, with the highest recall value of 99.43% among experimented models, shows its high sensitivity to detect COVID-19 cases, although it has a relatively bad classification ability in four-class classifying (accuracy of 93.34%). MViT2 and EfficientViT3, although maintaining moderate performance in the experimented four-class classification COVID-19 chest X-ray dataset, demonstrate outstanding accuracy in multi-class classification with other high-complex datasets [24,34].

For our model, which is based on ViT-Base-patch16 with specific adjustments, we remove unnecessary layers and retain crucial layers for feature extraction, reducing the model size. This, coupled with reduced trainable parameters (from 85.81 M to 71.62 M) and training time (from 313.94 s to 264.79 s), results in our model achieving the highest accuracy of 95.79% by mitigating overfitting. On the other hand, when delving deeper into the confusion matrix of the proposed model, as depicted in Fig. 5, we can discern the model's effectiveness. The most crucial class, COVID-19, achieves an accuracy of 98.58% based on $\frac{TP}{FN}$. However, the lung opacity and normal class exhibit suboptimal performance, which can be attributed to their relatively large image count and potential challenges posed by image quality in the classification

task. Thus, we hypothesize that the classification between lung opacity and normal images is the most challenging bias existing in this COVID-19 chest X-ray dataset. Additionally, by examining the ROC curves and AUCs (as shown in Fig. 5), we conclude that our proposed model is confident regarding the decision it makes, and an almost ideal ROC curve is plotted on the prediction of COVID-19 and pneumonia cases, with a COVID-19 AUC value of 0.9993, lung opacity AUC value of 0.988, normal AUC value of 0.9896, and pneumonia AUC value of 1. The slightly lower AUC value (0.988 and 0.9896) of lung opacity and normal classes can further prove their biases in our model training.

To comprehensively assess the advancements offered by our proposed model, we compare its accuracy and other metrics with existing research studies, as shown in Table 4. A major focus in the existing literature has been the binary classification of COVID-19 and non-COVID-19 cases. While such approaches achieve high accuracy (up to 97.83%), they lack the granularity needed for a more comprehensive diagnosis. Our multi-class classification approach addresses this limitation by focusing on identifying specific classes within the spectrum of COVID-19 and related conditions. Remarkably, our model achieved an outstanding accuracy of 99.57% when classifying the COVID-19 class. Furthermore, the high recall (98.58%) and leading specificity (99.76%) demonstrate the model's exceptional sensitivity in detecting COVID-19 features while accurately distinguishing other conditions. These results highlight the model's potential to improve the precision of medical diagnosis using chest X-ray images.

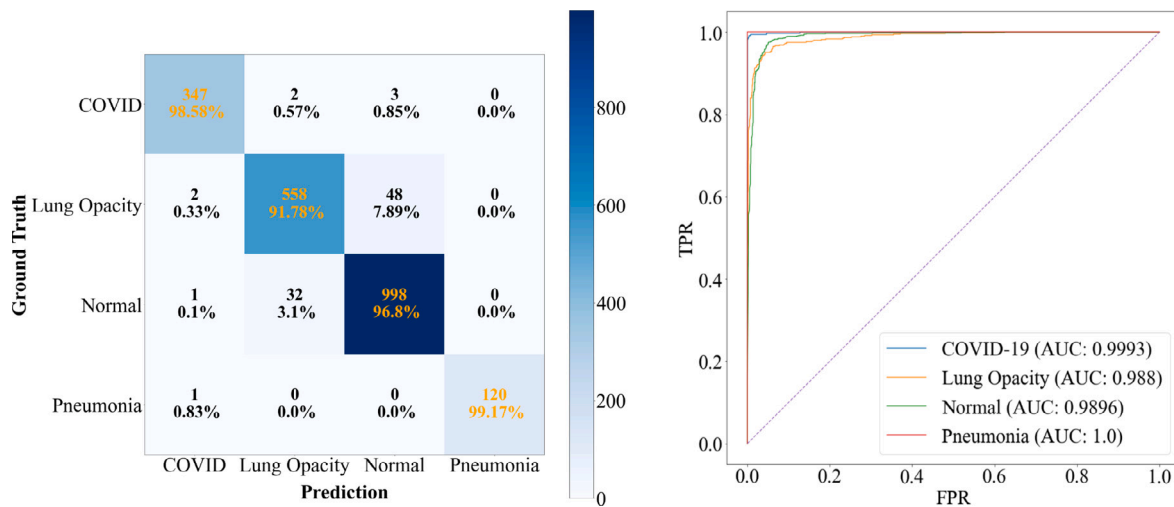
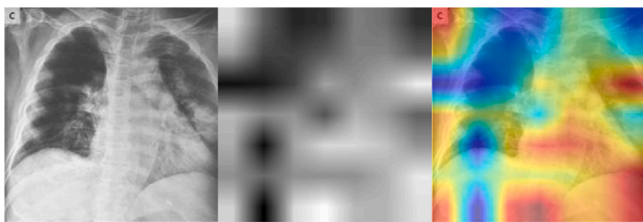
4.3. Results validation through visualization

In addition to evaluating the accuracy, we leverage the gradient class activation map (Grad-CAM) [42] to gain insights into our model's reasoning process when classifying COVID-19 chest X-ray images. Grad-CAM is a visualization technique that generates heatmaps highlighting the image regions most influential for specific class predictions. This can be particularly valuable for interpreting the model's decision-making process in the context of medical diagnosis. Grad-CAM works by intercepting logits in the feature extractor layer, along with a specific logit in the classifier layer corresponding to the predicted class. Backpropagating the gradients of these logits with respect to the logits in the extractor yields neuron importance weights. These weights quantify the contribution of each feature map to the final prediction. By combining these weights with the original features and applying a ReLU activation, Grad-CAM emphasizes only positive influences, resulting in a heatmap that visually pinpoints the areas crucial for the model's decision. In the actual validation case, the interception of feature logits and those regions are highlighted with white color and red color in gray and RGB images, respectively. Fig. 6 demonstrates a successful prediction in the COVID-19 image through our proposed model. The center region with fibrosis tissue highlighted in red indicates our model's attention, which matches medical diagnosis facts [43]. However, the red regions on the edge of the heatmap indicate the existence of biases within COVID-19 images, which reveals the possibility of further improvement by training models with more accurately segmented images. Grad-CAM not only helps verify the proposed model's performance but also ensures the appropriate layers of encoder while

Table 4

Comparison with existing studies (Test Result, Binary: COVID-19 vs. non-COVID, Three-Class: COVID-19 vs. Normal + Lung Opacity vs. Pneumonia).

Prediction model	Classification type	Accuracy (%)	Recall (%)	Precision (%)	F1 score (%)	Specificity (%)
Basic CNN [35]	Binary	93.99	89.16	95.63	87.41	88.28
Cycle GAN [36]	Binary	93.75	84.00	90.00	–	97.00
CovXmhc (VGG16 + SVM) [37]	Binary	95.00	95.00	96.00	95.00	96.00
Hybrid Deep Learning Mdoel [38]	Binary	92.00	92.00	93.00	92.00	98.68
Multi-Model [39]	Binary	97.83	95.45	100.00	97.67	100.00
DenseNet201 [28]	Three-class	95.11	94.55	94.56	94.53	95.59
DenseNet121 [40]	Three-class	93.5	92.59	93.44	93.00	–
Resnet50-BiLSTM [41]	Three-class	98.51	98.51	98.52	98.51	–
ViT-B32 [12]	Three-class	96.00	96.00	–	–	96.00
This work	Binary	99.57	98.58	98.87	98.73	99.76
	Three-class	99.57				
	Four-class	95.79				

**Fig. 5.** Confusion matrix (left) and ROC curves with AUC (right) for our proposed model.**Fig. 6.** Grad-CAM view from our proposed model on the COVID image. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

fine-tuning. Moreover, this heatmap generated based on our proposed prediction model can help doctors faster and more accurately diagnose. We believe that collaborating with medical experts to analyze Grad-CAM visualizations, particularly for negative predictions, could help uncover potential hidden biases within the model. This analysis could provide valuable insights for further refinement and potentially lead to improved accuracy.

5. Conclusion

This study investigates the effectiveness of computer vision models, particularly vision transformers, for diagnosing COVID-19 from chest X-rays. We meticulously analyze and compare the performance of

four diverse architectures: vision transformer, EfficientNet, MViT, and EfficientViT. We assess overfitting behavior in both CNN and VT models by visualizing the training and validation loss curves. Our analysis reveals several techniques to mitigate overfitting, with weight decay emerging as a particularly impactful method. Additionally, we explore the influence of varying the number of encoder blocks within the transformer models. Through careful fine-tuning and modifications, we present our proposed fine-tuned ViT model (FT-ViT) as a leading performer, achieving exceptional accuracy. FT-ViT demonstrates not only outstanding accuracy in four-class classification (95.79%) but also a remarkable 99.57% accuracy in a clinically relevant three-class grouping, along with consistently high performance in binary classification scenarios. Stringent validation using quantitative metrics, ROC curves, and Grad-CAM visualizations solidifies FT-ViT's effectiveness. Our validation process also identified potential biases within the dataset, particularly between lung opacity and normal images. We believe this challenge can be addressed through improved image segmentation and balancing techniques. This comparative analysis across various architectures highlights the superiority of our fine-tuned VT approach for COVID-19 diagnosis. We envision that FT-ViT, with its exceptional accuracy, could contribute to more efficient and potentially faster COVID-19 diagnosis, particularly as a screening tool in clinical settings. Furthermore, our findings provide valuable insights for future research in medical image analysis tasks. Beyond specific instructions, this work opens doors for exploring transfer learning from other medical imaging datasets to broaden the applicability of vision transformer-based models in the healthcare domain.

6. Future work

Throughout our research, we have encountered several intriguing findings that merit additional exploration and development. First, we observe that the performance of models trained on multi-level segmented images showed negligible improvement, and in some cases, a decline in effectiveness. This outcome suggests that the masks provided with the dataset might not adequately segment the images, leading to the loss of small yet critical details during the masking process. It raises the possibility that more sophisticated segmenting models, potentially transformer-based, could generate improved masks, enhancing model performance. Second, the EfficientViT [24] architecture displayed promising fitting curves, yet it struggled to converge with the dataset used in our experiments. This observation indicates the architecture's potential for diagnosing COVID-19 from chest X-ray images, suggesting that certain modifications – such as freezing the extractor layers and adapting the bottom linear layer to include trainable layers like the K-nearest neighbor algorithm – could facilitate further training and improve model convergence. Third, given the limitations imposed by the input dimension size of pre-trained models, our experiments were conducted using three-color channel images. We hypothesize that developing a model pre-trained on grayscale images could not only reduce the model size but also enhance prediction efficiency. Such advancements could extend the model's applicability to portable device applications, including those in the Internet of Things (IoT) domain [44]. Lastly, we recognize the potential of employing optimization techniques to elevate our models' accuracy and training efficiency. Similar approaches have demonstrated significant benefits in other machine learning contexts [45,46], suggesting a valuable avenue for future work to explore.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We use public datasets as described in [17,18,27–29].

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